



EGCI 232 Engineering Electronics

- Review on ECA



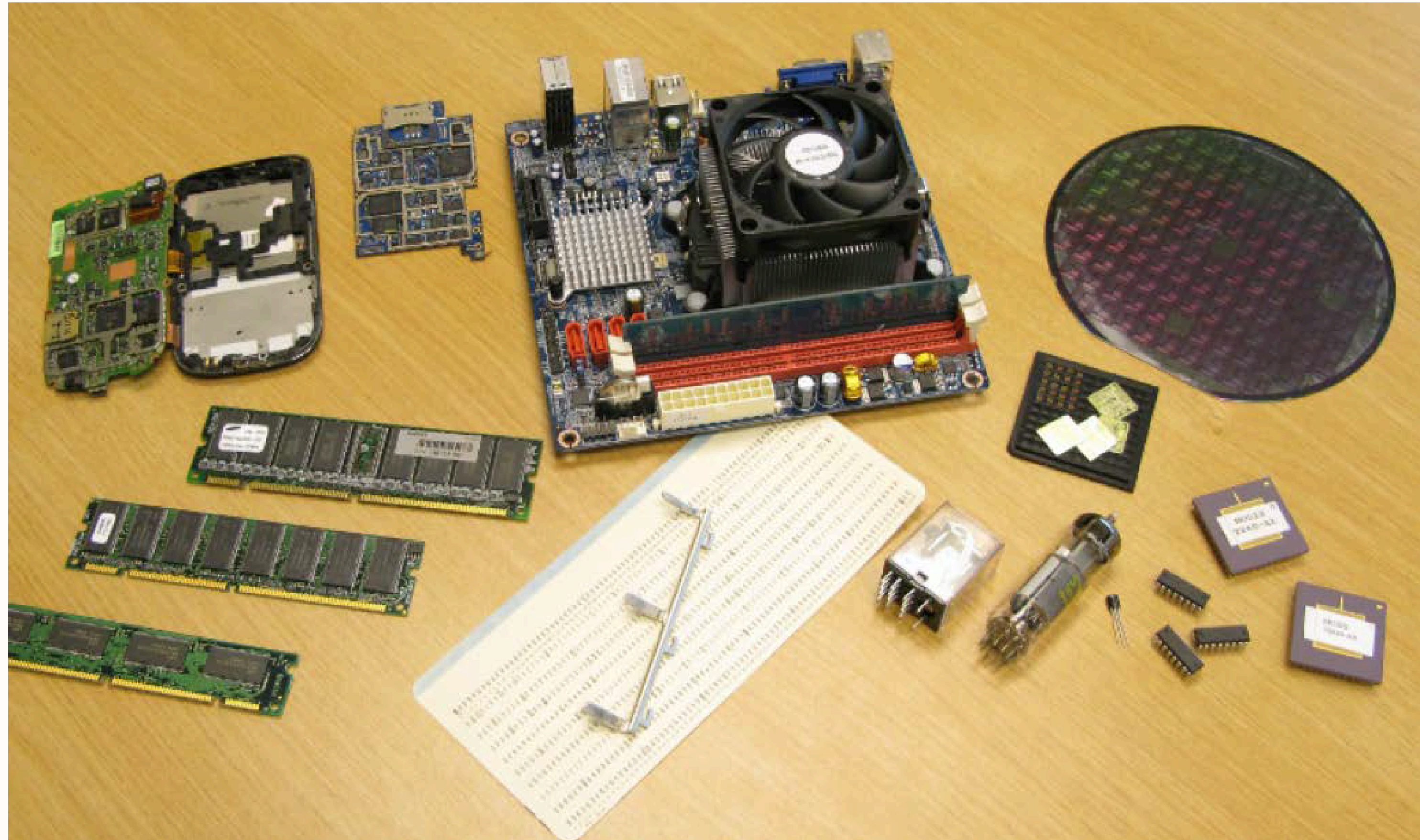
Let's join MS Team

EGCI 232 Trimester 1 AY 2026-2027

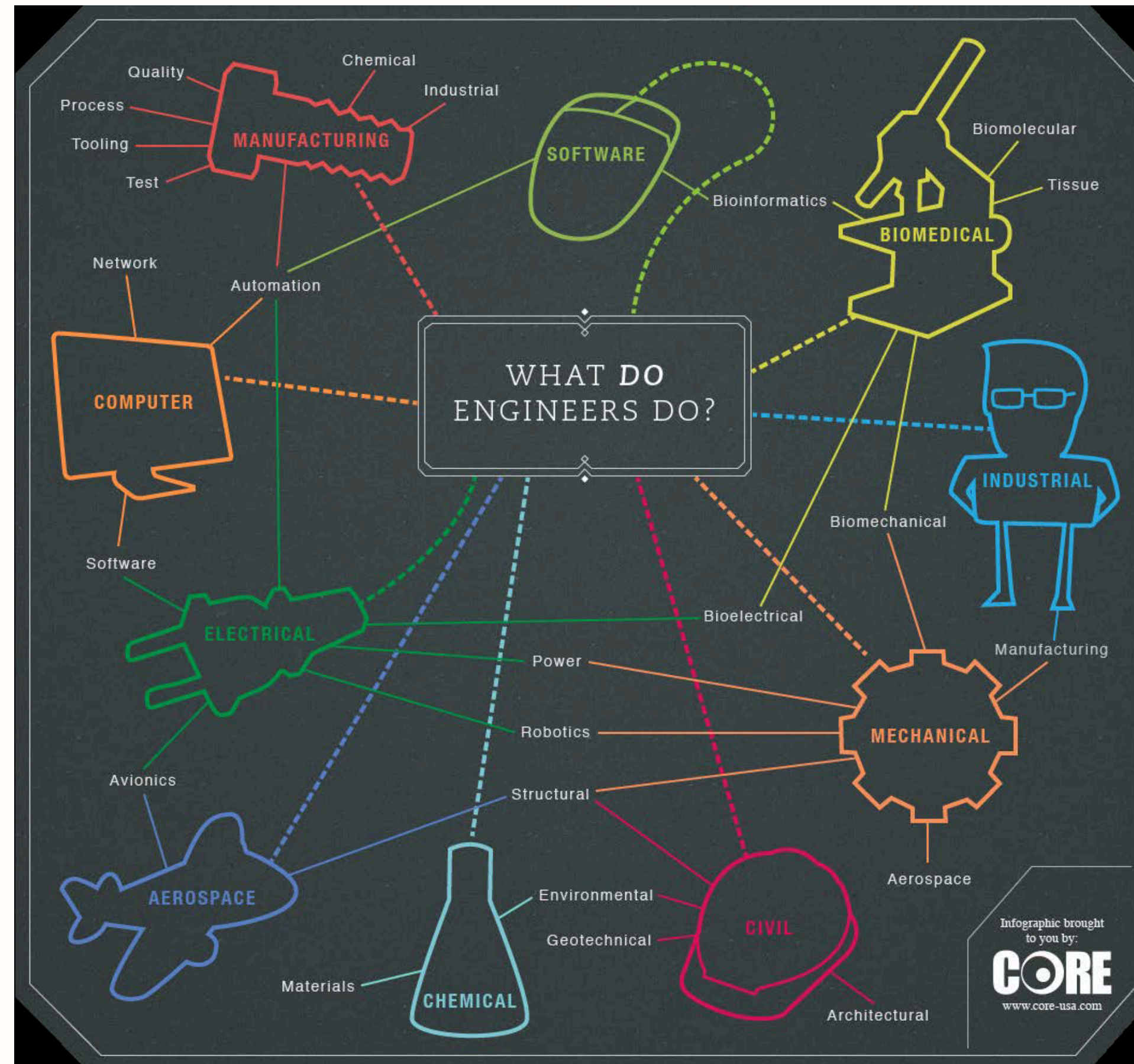
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What are we going to learn?





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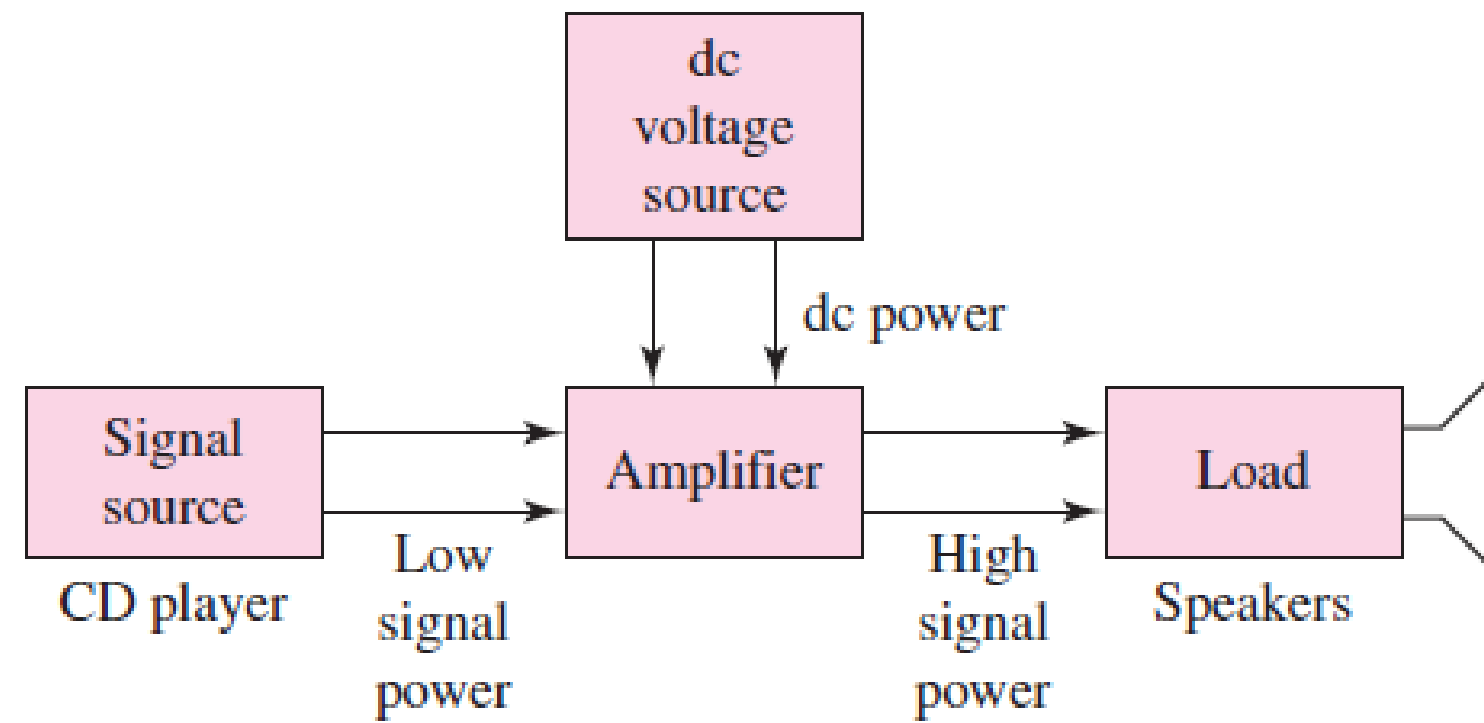


Course Outlines

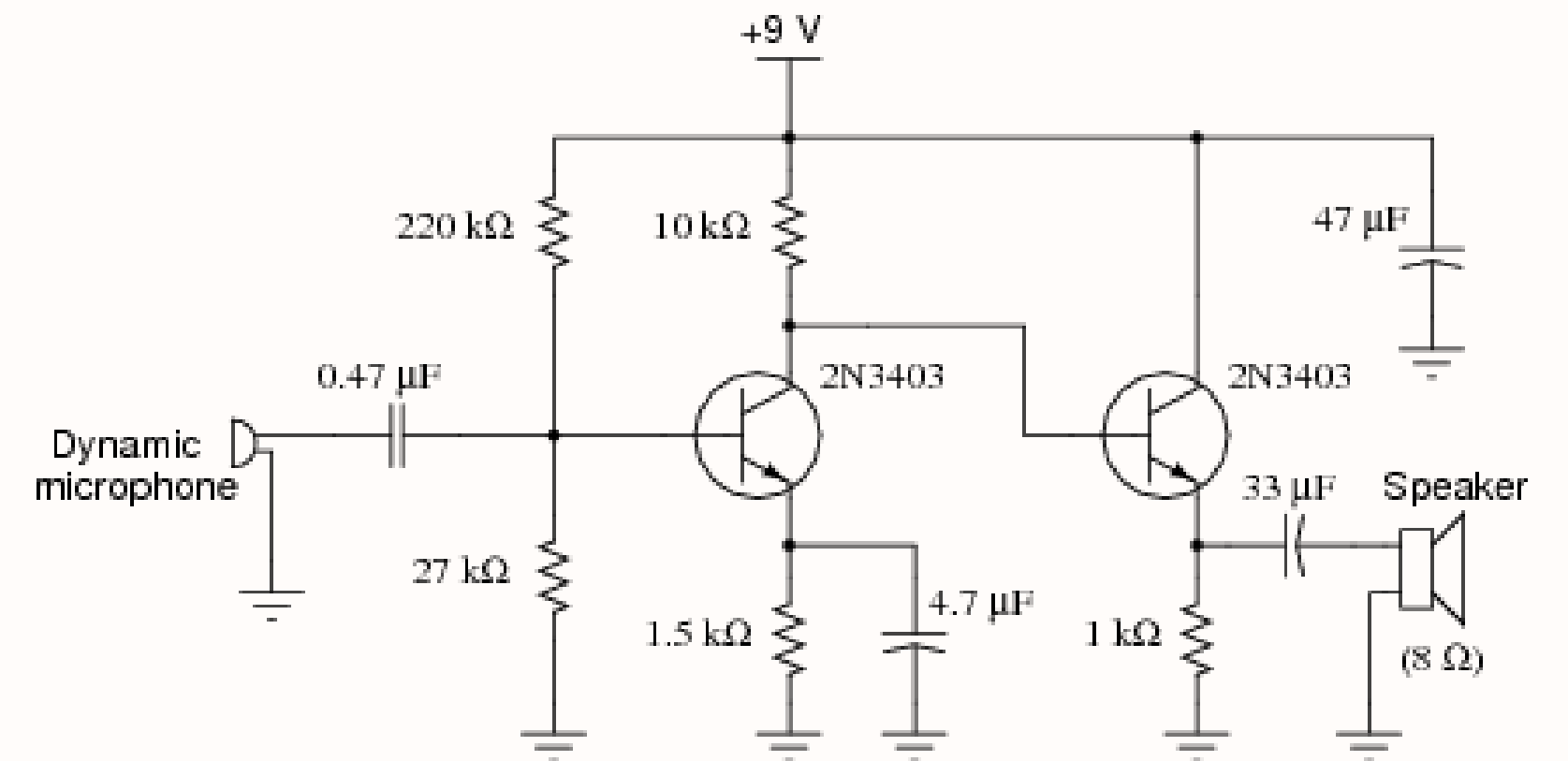
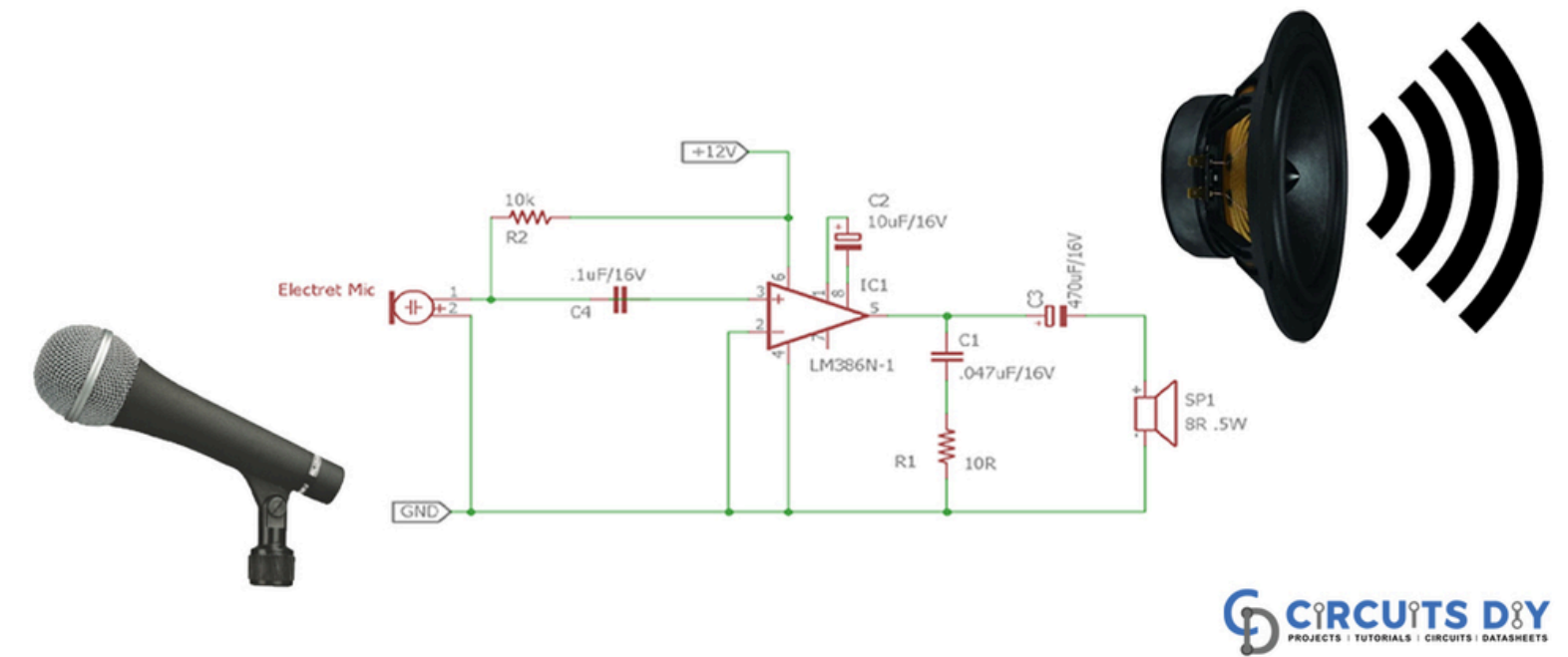
- Introduction to Electronics and Semiconductors
- Diode circuits
- Bipolar Junction Transistors (BJT) Amplifiers
- Field-Effect Transistors (FET) Amplifiers
- Digital Circuits
- Operational Amplifier (Op-Amp) Circuits and
- Applications



For Example



Microphone to Speaker Amplifier

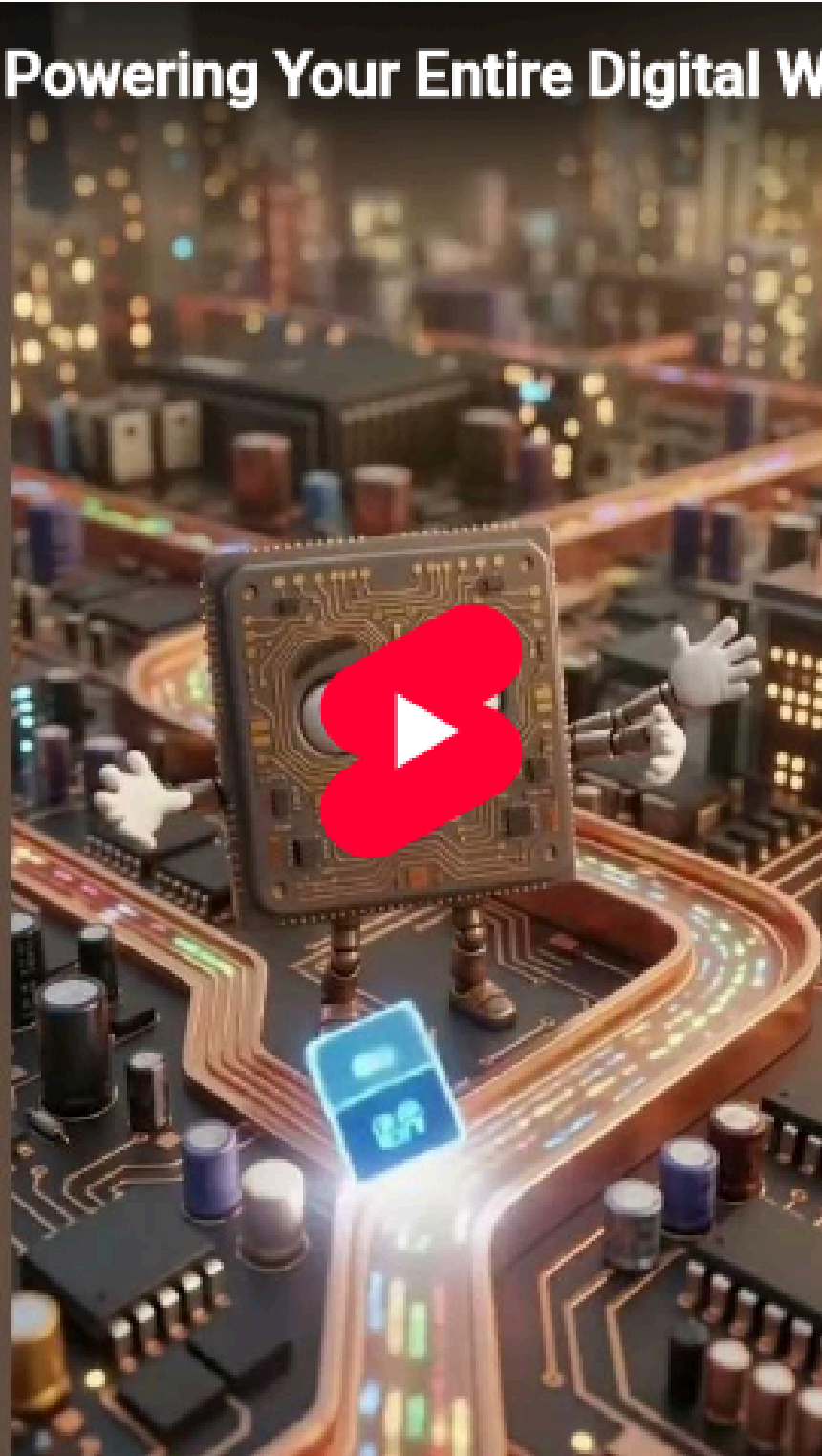




For Example



 **Inside the Microchip Powering Your Entire Digital World**
Educational Engineering Team



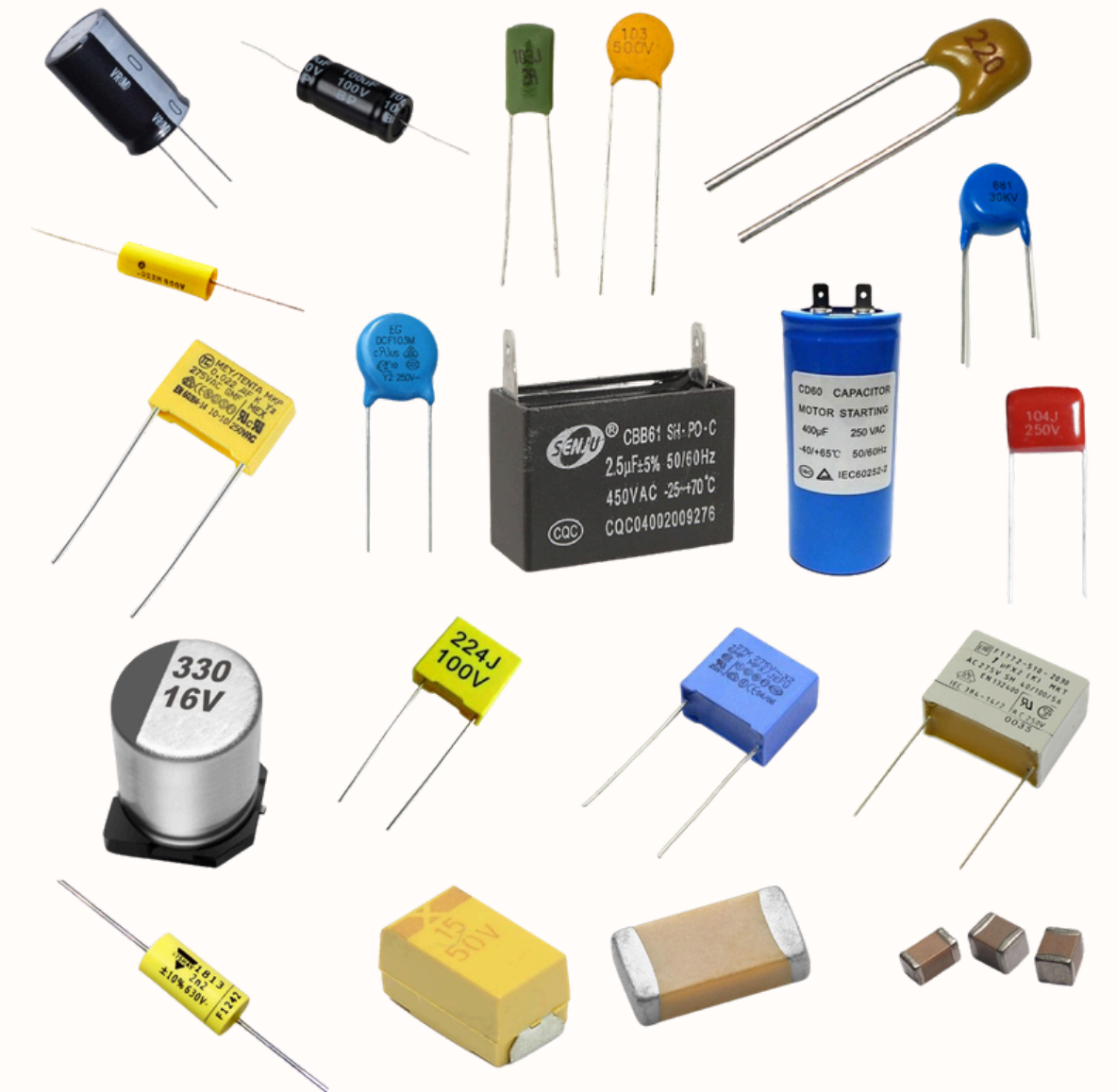
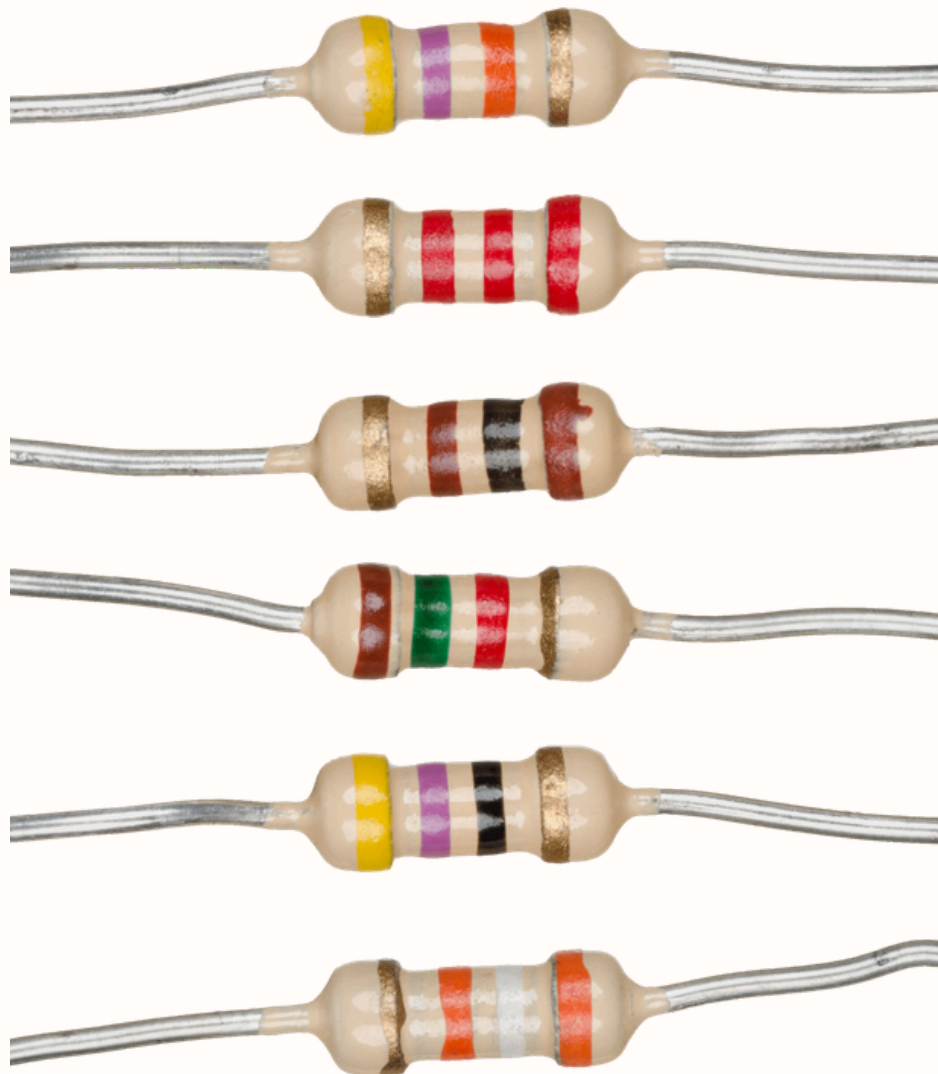


Basic Knowledge for the Course

- Fundamentals of Electrical Engineering
 - Electrical Devices
 - Ohm's Law
 - Kirchhoff's Laws
 - Thevenin's theorem
- Signals
- Physics of Elements



Fundamentals of EE





Fundamentals of EE



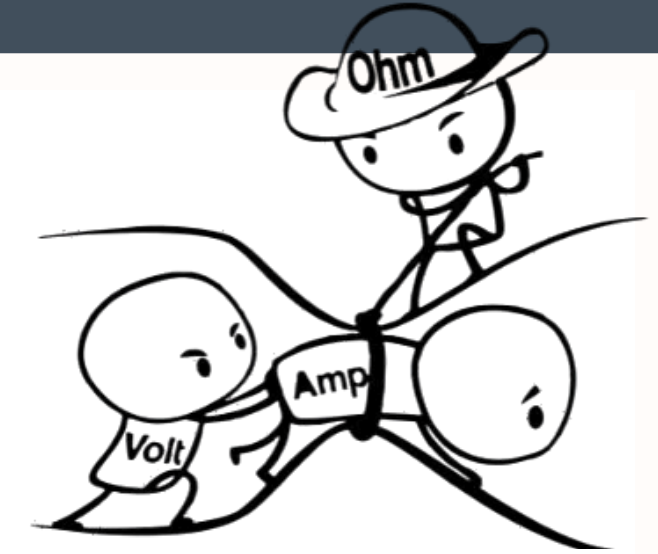
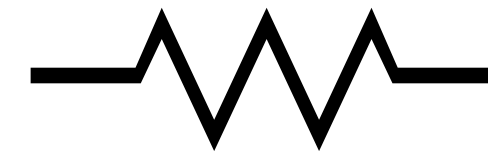


Fundamentals of EE

□ Ohm's Law

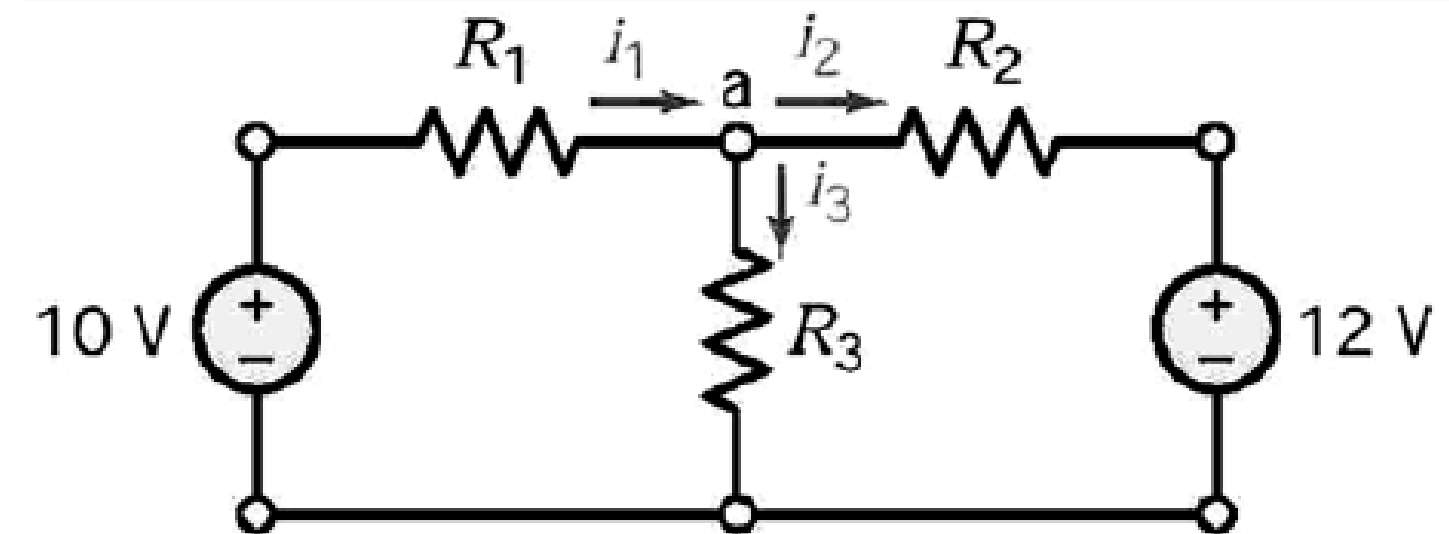
- ▣ Ratio between voltage and current is impedance

$$V = I \cdot R$$



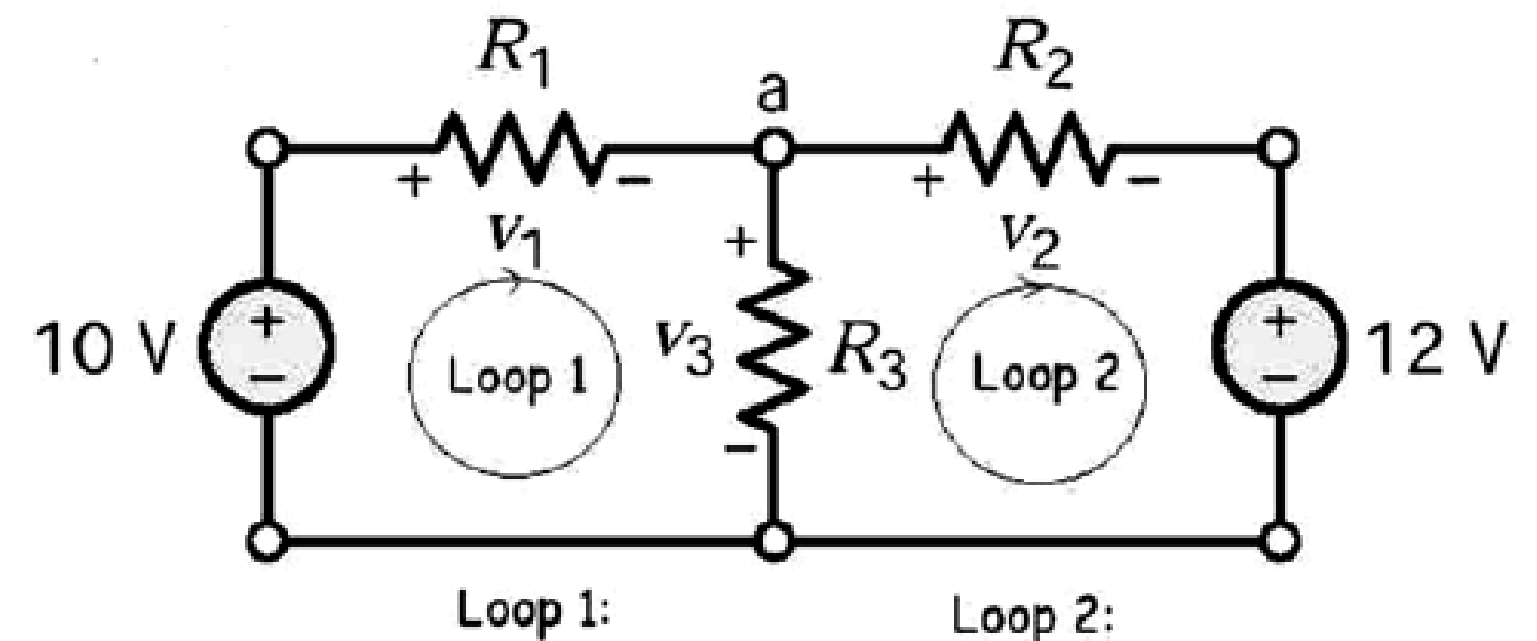
□ Kirchhoff's current law (KCL)

- ▣ The algebraic sum of currents entering any node is identically zero for all instants of time



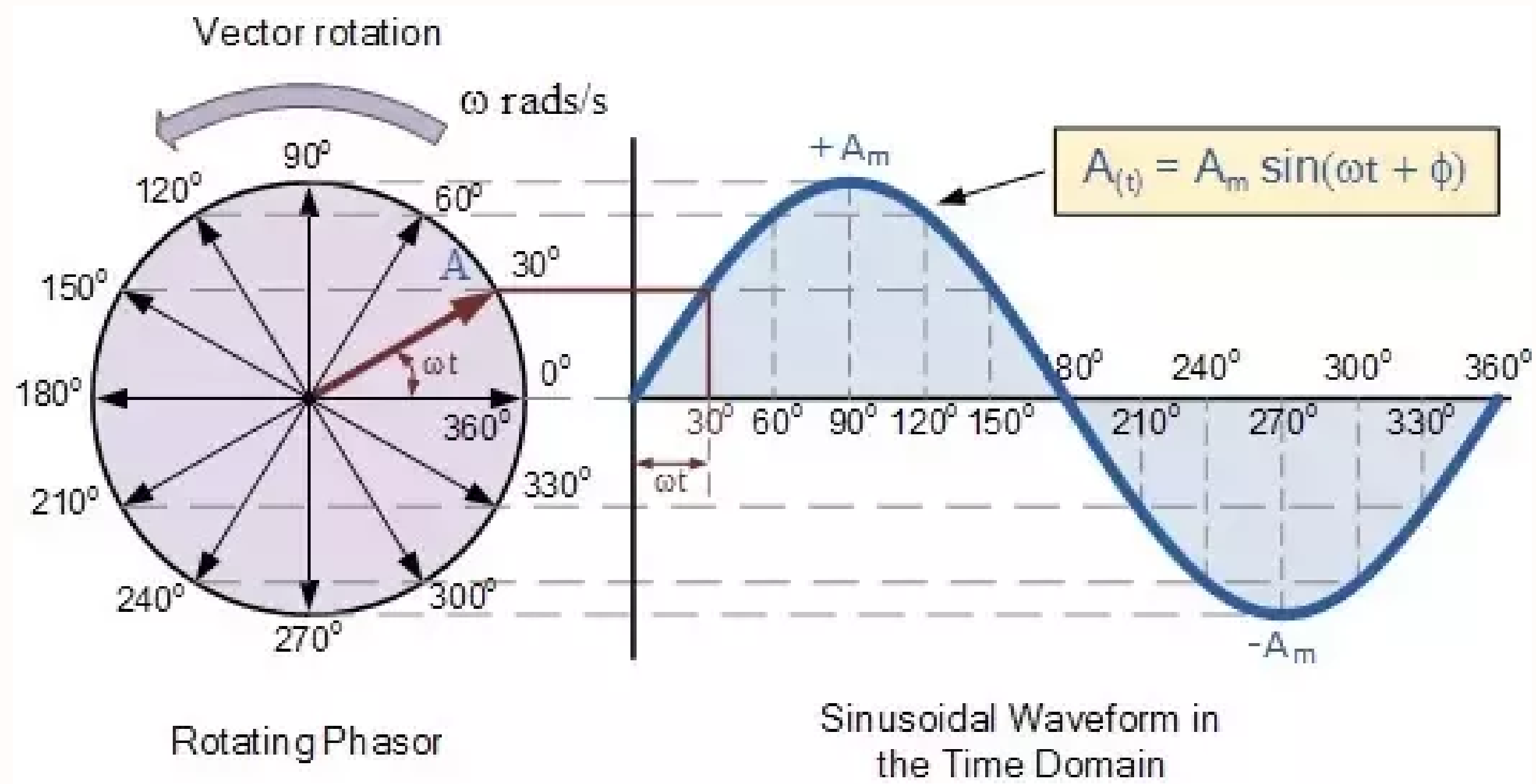
□ Kirchhoff's voltage law (KVL)

- ▣ The algebraic sum of voltages around any closed path in a circuit is identically zero for all time



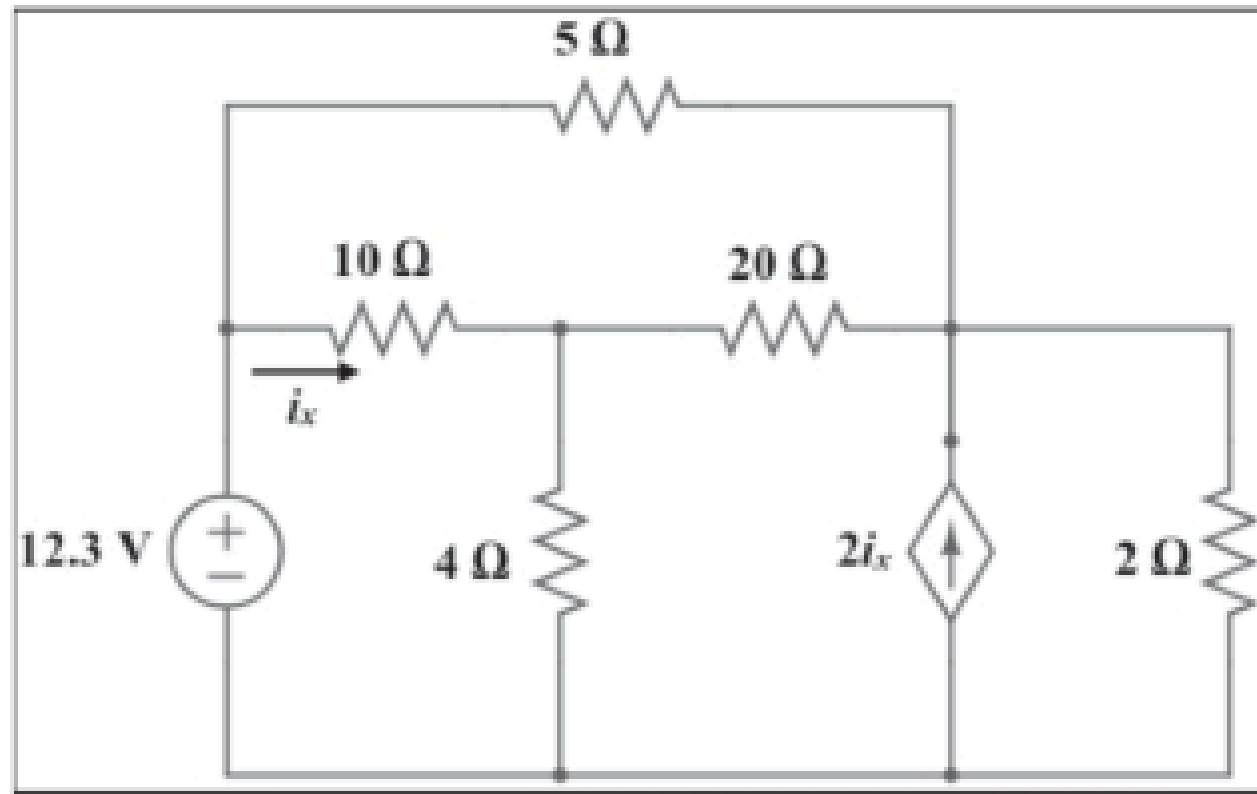


Signal





Fundamentals of EE





Fundamentals of EE

